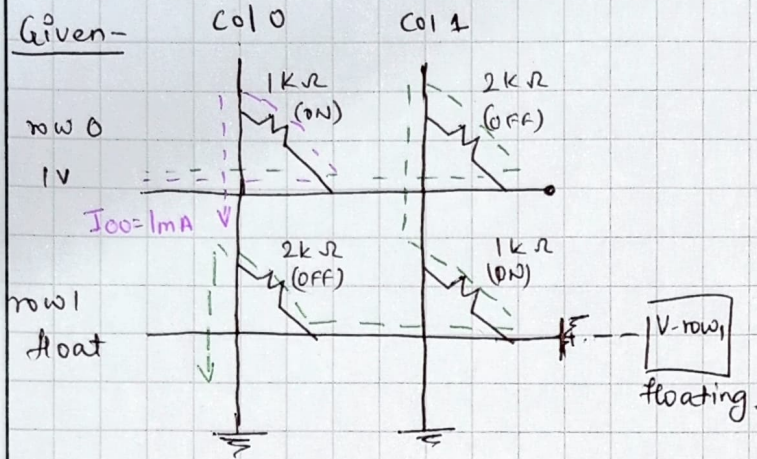


HARDWARE FOR AI-ML

ECE 510 CMAN 6

Given-



Task

1) To find - Ideal current output current $I_{\text{col } 0}$

→ Solution -

Ohm's Law $V = IR$ $(V = 1V, R = 1k\Omega)$

$$\therefore I_{00} = \frac{1V}{1k\Omega}$$

$$\therefore I_{00} = 1mA$$

2)

→ Solution KCL Sneakpath.

KCL at $V_{\text{row } 1}$

$$\left(\frac{V_{\text{row } 1} - 0}{2k} \right) + \left(\frac{V_{\text{row } 1} - V_{\text{col } 1}}{1k} \right) = 0$$

$$\frac{V_{\text{row } 1}}{2} + V_{\text{row } 1} - V_{\text{col } 1} = 0$$

$$\frac{3}{2} V_{\text{row } 1} = V_{\text{col } 1} \quad \text{--- (1)}$$

KCL at $V_{\text{col } 1}$

$$\left(\frac{1 - V_{\text{col } 1}}{2k} \right) + \frac{V_{\text{row } 1} - V_{\text{col } 1}}{1k} = 0$$

$$\frac{1 - V_{\text{col } 1}}{2} + V_{\text{row } 1} - V_{\text{col } 1} = 0$$

$$\therefore \frac{1}{2} + V_{\text{row } 1} - \frac{3}{2} V_{\text{col } 1} = 0 \quad \text{--- (2)}$$

$$\frac{1}{2} + V_{\text{row } 1} - \left(\frac{3}{2} \right) \left(\frac{3}{2} \right) V_{\text{row } 1}$$

$$\therefore V_{\text{row } 1} = \frac{2}{5} = 0.4V$$

$$V_{\text{col } 1} = \frac{3}{5} = 0.6V$$

I_{col 0} component = $I_{00} + I_{01}$

$$= 1mA + 0.2 = 1.2mA$$

Error = 20%

$$= \frac{0.4 - 0}{2k} = 0.2mA$$

3) → The sneak path ($R_{01} \rightarrow R_{11} \rightarrow R_{10}$) creates a parasitic current path adding 0.2mA of spurious current. This corrupts the dot product output, in large arrays the number of parasitic paths grows combinatorially, causing severe errors in MMV results.